Customer Churn Data Analytics Project

Stage 5 Summary: Data Cleaning & Preparation

This document summarizes the Data Cleaning & Preparation stage of the Customer Churn Data Analytics Project, based on the notebook *05. Data Cleaning & Preparation*. The goal of this stage is to transform the raw dataset into a clean, consistent, analysis-ready dataset, using the quality issues identified in Stage 4.

> **Guiding principle: every cleaning action should be justified by a finding from Stage 4 and should be reproducible. Unlike Stages 3 and 4, this stage intentionally modifies the working data — but the original raw dataset is never overwritten.**

1. Stage Activities

Step	Activity	Description
5.1	Load Raw Dataset	Load the untouched raw dataset as the starting point for cleaning.
5.2	Create a Working Copy	Create a separate DataFrame for cleaning; preserve the raw dataset unchanged.
5.3	Apply Cleaning Rules	Apply only the cleaning decisions justified by Stage 4.
5.4	Handle Missing / Blank Values	Convert identified blank representations to proper missing values and decide how to handle them.
5.5	Correct Data Types	Convert variables to their appropriate analytical data types.
5.6	Standardize Categorical Values	Remove unnecessary whitespace and standardize confirmed inconsistent category labels.
5.7	Handle Duplicate Records	Remove or retain duplicates according to findings and documented business rules.
5.8	Validate Customer Uniqueness	Ensure the final customer-level dataset has the intended grain and identifier structure.
5.9	Handle Invalid Values	Correct, replace, or remove confirmed invalid values according to documented rules.
5.10	Handle Outliers	Decide whether potential outliers are valid observations or actual errors; do not remove legitimate customers simply for being statistically unusual.
5.11	Create Derived Variables	Create justified analytical variables such as churn_flag, senior_citizen_label, and tenure groups.
5.12	Validate Relationships	Re-run key quality checks after cleaning to confirm transformations did not introduce new problems.
5.13	Compare Before vs. After	Compare row count, columns, missing values, duplicates, data types, and key distributions.
5.14	Save Clean Dataset	Export the final analysis-ready dataset to a separate file.
5.15	Document Cleaning Log	Record each issue, action taken, reason, and effect on the dataset.

2. Cleaning Workflow (Notebook Overview)

Setup & Working Copy

The raw CSV is loaded into raw_df, which remains the untouched source of truth. A separate working copy, clean_df, is created via raw_df.copy() and used for every subsequent transformation. The final cleaned dataset is exported to a distinct file path/directory, never overwriting the raw source.

Column & Text Standardization

- Column names are converted to a consistent snake_case convention (e.g., CustomerID → customer_id, MonthlyCharges → monthly_charges, TotalCharges → total_charges), and all downstream code consistently references the new names.
- Leading/trailing whitespace is stripped from all object/string columns, without altering meaningful category wording such as "No internet service".
- Standardized categorical values are validated by inspecting value_counts() after trimming, rather than applying a blind remapping that could distort valid labels.

Data Type Correction

tenure, monthly_charges, and total_charges are explicitly converted to numeric via pd.to_numeric(errors="coerce"), since total_charges may contain blank strings that prevent direct numeric interpretation.

Handling Missing total_charges

Records with missing total_charges after conversion are inspected against tenure, monthly_charges, contract, and churn before any treatment is applied. The documented business rule — that these missing values are associated with zero-tenure customers — is explicitly verified: if any missing record does not have zero tenure, the notebook raises an error and stops rather than silently filling the value. Only after this check passes are the missing values filled with 0, consistent with the business interpretation of accumulated charges for brand-new customers.

Duplicates & Uniqueness

- Exact duplicate rows are counted first; rows are dropped only if duplicates are actually present, avoiding unnecessary changes when the count is zero.
- customer_id uniqueness is validated (missing IDs, unique ID count vs. row count, duplicate ID count); any duplicate customer IDs are inspected in full rather than dropped automatically, preserving the intended customer-level grain.

Numeric Range Validation

Negative values in tenure, monthly_charges, and total_charges are checked explicitly. If any impossible negative value is found, the notebook raises an error and stops for investigation rather than silently converting or removing the value. Outliers are not removed simply for being statistically large.

Derived Variables

- senior_citizen_label: a business-friendly Yes/No label mapped from the original 0/1 seniorcitizen field, with the original field retained.
- churn_flag: a numeric 1/0 flag derived from the churn text field (Yes → 1, No → 0), after verifying no unexpected churn values exist.
- tenure_group: a four-band segmentation (0–12, 13–24, 25–48, 49+ months) created with pd.cut() for churn comparisons, while the original tenure value is preserved.

- monthly_charge_band: a three-level Low/Medium/High band created with pd.qcut() using tertiles, with duplicates="drop" to avoid failures from identical quantile boundaries.
- high_monthly_charge: a boolean flag marking customers at or above the 75th percentile of monthly_charges, created purely as a reusable analytical segment rather than a risk conclusion.

Post-Cleaning Validation

After all transformations, the main quality checks are re-run on the cleaned dataset: shape, total missing cells, duplicate rows, duplicate customer IDs, per-column missing counts, and data types. A final describe() summary confirms that seniorcitizen, tenure, monthly_charges, total_charges, and churn_flag are genuinely numeric and have plausible distributions.

Export

The cleaned, analysis-ready dataset is saved as a separate CSV file (telco_churn_cleaned.csv) in a dedicated "Cleaned" directory, keeping the raw source file completely untouched.

3. Cleaning Log

The table below documents every meaningful cleaning decision made in this stage, the action taken, and the business/technical reason behind it.

Issue	Action	Reason
Text whitespace	Stripped	Prevent accidental duplicate categories
Numeric fields stored as text	Converted to numeric	Required for analysis
Blank TotalCharges	Converted to missing, validated, then filled with 0 for zero-tenure records	Consistent with the business interpretation of accumulated charges
Exact duplicate rows	Removed only if present	Prevent double counting
Duplicate customer IDs	Investigated, not blindly removed	Customer-level grain must be preserved
SeniorCitizen	Added readable label	Better business communication
Churn	Added numeric flag (churn_flag)	Easier analytical calculations
Tenure	Added tenure groups (tenure_group)	Segment analysis
Monthly charges	Added bands/flag (monthly_charge_band, high_monthly_charge)	Later diagnostic analysis

The final presentation and reporting should use the actual counts produced by running this notebook against the dataset, rather than hard-coded or assumed numbers.

4. Deliverable

A validated, analysis-ready cleaned dataset (telco_churn_cleaned.csv) plus a documented cleaning log recording every issue addressed, the action taken, and the reasoning, ready to support the subsequent analysis and reporting stages.

Prepared as part of the Customer Churn Data Analytics Project — Data Cleaning & Preparation stage.